

Section 2

IDENTIFYING CRITICAL SKILLS GAPS

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PRE-SKILLING YOUR WORKFORCE FOR AN UNPREDICTABLE FUTURE

by Tomas Chamorro-Premuzic

Despite the many clickbaity headlines and apocalyptic fearmongering around AI and job automation, the data is rather clear: As I highlight in my latest book, *I, Human: AI, Automation, and the Quest to Reclaim What Makes Us Unique*, you are less likely to lose your job to AI than to another human using AI, especially if you don't use AI yourself. In that sense, there's nothing new about the AI age; just like previous disruptive technologies, AI is eliminating some jobs but creating many more new jobs in turn, which desperately require humans.

As illustrated in our recent ManpowerGroup report, 58% of employers see AI as a net creator of new jobs.¹ The problem is that people who are displaced by job automation (for example, brick-and-mortar store managers) don't automatically have access to all the new jobs that are created by technology (for example, cybersecurity analyst, digital marketer, or AI ethicist). According to the World Economic Forum, half of employees will need reskilling by 2025 to keep up with new technologies, and this figure predates the current generative AI boom.²

As Stanford's Erik Brynjolfsson estimates, for every dollar organizations spend on technology, they need to invest an additional nine dollars in talent and related processes, such as identifying people with the right skill sets and potential, developing the right skills, enabling proper change-management processes, and creating the conditions for people to live up to their potential.³ Indeed, as my colleague Becky Frankiewicz and I noted, what makes or breaks digital transformation is not the actual technology, but the human factor: talent.⁴

So, what we need is not just reskilling or upskilling, but *pre*-skilling—that is, being able to future-proof talent and reinvent peoples' careers before we even know what tomorrow's jobs and in-demand skills will be.

Pre-Skilling Your Workforce for an Unknown Future

How can this be done? Although nobody has the exact recipe, not least because there's no data on the future, we can extrapolate from macro historical trends, as well as draw lessons from recent economic and human capital patterns, to make five broad recommendations:

1. Focus on potential

As the lifespan of current skills, expertise, and performance shrinks, it is advisable to hire and promote people for what they *could* do, rather than what they *have done* in the past. This means prioritizing soft skills—such as learning ability, curiosity, resilience, and adaptability—over hard skills—such as programming or data mining, focusing on the foundational ingredients of employability rather than current technical expertise or past employment history, both of which still dominate modern hiring approaches.

Simply put, we may not know what tomorrow's jobs will look like, but we can safely assume that when people are more curious, emotionally intelligent, resilient, driven, and intelligent, they will generally be better equipped to learn what is needed to perform those jobs and provide whatever human value technology cannot replace. One advantage to potential is that we have reliable and proven methods to measure it, notably psychological assessment tools, which can also be used to *develop* people's potential, particularly through normative and data-driven feedback that helps them identify gaps between where they are and where they need to be.⁵

2. Provide crucial feedback

Even before AI's arrival, there was already a great deal of confusion and misunderstanding around aligning people's talents and potential with the best career choices. Things are even more complex now, and most employees (and managers) are rightly wondering what they will do in the future, whether their experience and expertise—and all the time and money they invested in their careers—will still be relevant, and if so, where.

Sharing data-driven feedback (from assessments, internal data, and peer ratings) and helping people understand how their interests and skills can be future assets to your organization are key. This means thinking of leaders as coaches or talent agents, where their main job is to proactively nurture people's talents as well as harness their full potential. Remember: Most people are feedback-deprived, and a staggering two-thirds of feedback interventions fail to produce desirable results. The key is to tell people what they *need* to hear even when they don't *want* to hear it and above all, to tell them something they didn't know and that can motivate them to get better.

Crucially, it is not enough just to provide people with feedback on their skills gaps; organizations also need to incentivize them to develop the relevant skills while providing the most efficient, enticing, and impactful programs to develop them.

3. Focus on talent expansion

Rather than betting on specialists or forcing people into specific niches, focus on broadening or expanding people's talents. This means *not* playing to people's strengths but, rather, helping them develop *new* strengths, so they become more versatile versions of themselves.

We all have a natural tendency to adapt the world to our current skills so that we can do “more with less” and optimize for lazy efficiency rather than stretch goals or new skills acquisition. In reality, unless

your employees are in the top 1% of any area of expertise (which, by definition, can only apply to a very small selection of individuals), they will be better served becoming jacks of all trades, not least because with every new task or skill you learn, there will be many new underlying skills with adjacencies to other tasks, jobs, and careers.

This will also increase diversity and inclusion. As Harvard's Joseph Fuller and Christina Langer and Burning Glass Institute's Matt Sigelman point out: "The shift to skills-based hiring will open opportunities to a large population of potential employees who in recent years have often been excluded from consideration because of degree inflation."⁶ If organizations complain about their inability to find diverse and creative talent, they should stop looking for talent in the same old places and ways.

4. Invest in mid-level managers

As Amy Edmondson and I recently noted, managers hold the key to unlocking human potential at work, especially when the challenge is to revitalize, reenergize, and reimagine talent.⁷ Even before the AI age, the role of middle managers was critical and accounted for 30% to 40% of team performance, but the role has increased massively in complexity.⁸

Historically, a manager was someone with proven technical expertise and strong past performance as an individual contributor. Today, they need to understand not just the "classic" managerial tasks (hire the right people, give them the right tasks and direction, use feedback to motivate them, and enable team performance), but also some very complex specific new challenges, such as managing hybrid and virtual teams, creating psychological safety, boosting DEIB (which includes hiring and motivating people who are different from themselves), and helping people manage the human-AI age, including all the uncertainties and stress that evokes.

In short, there has never been a better time to invest in mid-level managers, especially harnessing the critical soft skills that will likely future-proof them by enabling them to keep up with technological advancements and drive the value realization their organizations expect from new innovations.

5. Invest in leadership skills

If you are worried about an AI-fueled future but still think humans will be part of the picture, then you should be worried about pre-skilling your leaders, for they will (still) be tasked with setting and selling the company's strategy, and driving the evolution of its culture, in tomorrow's uncertain time. In part, this is still about promoting individuals with the right soft skills (as per step 1) and developing those skills. It is also about moving on and understanding that today's top leaders may be a poor fit for the future, not least because their success is based on replicating what has worked in the past, which is often an obstacle to changing or learning new things.

Importantly, investing in leadership means harnessing the skills that enable people to collaborate effectively and become a high-performing team. It is not about promoting individuals to senior positions or glorifying status, let alone about the popularity contest that often underpins "high-potential" nominations—rather, it is about the qualities people must have to inspire, motivate, and bring others along into a future that will surely include AI but should ideally also be defined as the human-AI age. This is the critical leadership challenge of today. If organizations can upgrade their leadership-selection systems, shifting from replicating past success profiles to selecting and nurturing for future skills, all other challenges will become easier. In particular, this means deemphasizing things like "culture fit,"

past performance, and technical expertise to focus more on learning ability, curiosity, integrity, and people skills. More importantly, if organizations are truly interested in harnessing diversity, they must prioritize inclusive leadership and cognitive diversity, breaking the homogeneity in leadership profiles.

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Technology is always changing, but to ensure that those changes are actually an evolution (i.e., contributing to progress) we must make every possible effort to enable humans to develop and deploy the relevant talents to reap the potential benefits of technology. Although this is by no means a given, we should have faith and confidence in our species' ability to continue to adapt to its own technological inventions.

TAKEAWAYS

We may not know what tomorrow's jobs will look like, but we can assume that individuals who are curious, emotionally intelligent, resilient, driven, and intelligent will be better equipped to learn and provide value that technology cannot replace. Organizations should focus on expanding talents and *pre*-skilling.

- # Focus on potential. Hire and promote individuals based on their future capabilities rather than past performance, given the decreasing lifespan of current skills.
- # Provide crucial feedback. Offer honest assessments of employees that motivate them to improve, even when it may be difficult for them to hear.
- # Focus on talent expansion. Prioritize broadening employees' skills and abilities over forcing them into specific roles.
- # Invest in mid-level managers. They are essential for unlocking human potential and revitalizing talent.
- # Invest in leadership skills. Developing leadership abilities in all employees fosters effective collaboration and enhances team performance.

NOTES

1. Manpower Group, "58% of Employers Believe Artificial Intelligence and Virtual Reality Will Be Job Creators," press release, Manpower Group, June 8, 2023, www.manpowergroup.com/en/news-releases/news/58-of-employers-believe-artificial-intelligence-and-virtual-reality-will-be-job-creators.
2. L. Li, "Reskilling and Upskilling the Future-ready Workforce for Industry 4.0 and Beyond," *Information Systems Frontiers* (2022), <https://doi.org/10.1007/s10796-022-10308-y>.
3. Erik Brynjolfsson, "The AI Awakening: Implications for the Economy," July 23, 2020, video, 1:00:48, www.youtube.com/watch?v=LNx_6gzLIvI.
4. Becky Frankiewicz and Tomas Chamorro-Premuzic, "Digital Transformation Is About Talent, Not Technology," hbr.org, May 6, 2020, <https://hbr.org/2020/05/digital-transformation-is-about-talent-not-technology>.
5. Tomas Chamorro-Premuzic, *The Talent Delusion: Why Data, Not Intuition, Is the Key to Unlocking Human Potential* (London: Piatkus, 2017).
6. Joseph Fuller, Christina Langer, and Matt Sigelman, "Skills-Based Hiring Is on the Rise," hbr.org, February 11, 2022, <https://hbr.org/2022/02/skills-based-hiring-is-on-the-rise>.
7. Amy C. Edmondson and Tomas Chamorro-Premuzic, "3 Ways to Reenergize Your Team When Morale Is Low," hbr.org, July 14, 2023, <https://hbr.org/2023/07/3-ways-to-reenergize-your-team-when-morale-is-low>.

8. R. Hogan and R. B. Kaiser, “What We Know About Leadership,” *Review of General Psychology* 9, no. 2 (2005): 169–180, <https://doi.org/10.1037/1089-2680.9.2.169>.

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THE SKILLS YOUR EMPLOYEES NEED TO WORK EFFECTIVELY WITH AI

by Nada R. Sanders and John D. Wood

Right now, the corporate world is in the throes of a generative AI hype cycle. Leaders have been inundated with speculation about the technology's potential to transform work. However, this raises an entirely new set of problems for leaders: They now understand that the future will involve humans working with AI and feel immense pressure to “do something” to implement AI solutions. But how do they actually integrate AI with their workforce to achieve good business outcomes?

Since the launch of generative AI tools such as ChatGPT in 2022 and the tsunami of other applications that have followed, we have conducted dozens of in-depth CEO interviews across industry sectors. Our goal was to identify how generative AI has changed the practices of leading firms and extract lessons for success.

Our recent interviews reveal a marked shift in focus from earlier studies—both our own and those of other researchers. While our own earlier interviews showed the importance of a human-focused culture, our recent interviews have signaled a definitive shift toward development of individual human competencies.¹ The current hype cycle is around AI capability, digital skills, and the need for organizational change management, but our current finding reveals the importance of specific human capabilities to the successful use of AI.

In fact, it is the human ability to understand context—which AI tools lack—that necessitates the need for greater human skills. This is an important insight that can help leaders understand the necessary human elements that drive an organization's successful use of AI.

The Skills Companies Really Need

We observed two categories of human skills that leaders see as critical—and which organizations are actually teaching to their employees. First are effective interpersonal skills, such as basic conflict resolution, communication, skills of disconnecting from emotions, and even mindfulness practices.

Second is domain expertise, with a focus on preserving that knowledge among experienced talent and developing it among young, inexperienced workers.

While digital literacy was viewed as a given by the leaders we spoke to, interpersonal skills such as the ability to effectively communicate, meaningfully engage with others, and garner team cooperation were not. Our interviews reveal that while technical skills get more press, it is these uniquely human skills that companies need—and find to be in short supply in today’s marketplace. Peter Cameron, CEO of Lenox Group, underscored that workplace success depends on the ability to cultivate personal relationships, saying that, “Nothing replaces long-term relationships that are personal—and the longer the relationship the better.” Maria Villablanca, cofounder and CEO of Future Insight Network, pointed to specific qualities to look for in talent: “People that can be creative and innovative in the way they find solutions—problem solvers.” Broader research backs this up: A study of 1,700 global companies found that companies that excelled on human capital metrics were four times as likely to have superior financial performance.²

One of the greatest values of experienced workers is domain expertise—deep knowledge of one’s environment. As AI takes over more tasks, there is a significant danger of atrophy of skills and loss of this kind of knowledge. John Sicard, president and CEO of the supply chain management company Kinaxis, pointed out that domain knowledge is essential to helping a business navigate through turbulent times. He gave the analogy that when an airplane experiences anything unusual, the autopilot is immediately turned off and pilots take control—at which point the pilots need to know what to do. Developing and maintaining these skills is essential.

Further, generative AI is shown to be more useful as a copilot for senior employees who can sift through AI “hallucinations”—inaccurate information presented as fact—and take the output as an aid. Inexperienced employees, however, may not be discerning enough and need a path to develop this knowledge. The sentiment was echoed by Ted English, former CEO of TJX Companies and current executive chairman of Bob’s Discount Furniture, who told us leadership requires “a lot of instinct, experience, and knowledge. Some of it you can’t get from a machine. Technology reinforces and allows you to make a more confident decision.”

While much has been written on the need to combine humans with AI, the real issue is how to implement this in practice. Here’s what companies need to do.

More Than Just Combining Humans and AI

In 2020 we published our Four I framework for using AI that calls for creating a human-centric organization.³ As a result of our current findings, we have modified the descriptors of our original framework, making it more specific in its mandates. This offers a direct comparison—and shift—between the use of AI in the post-pandemic business environment.

The first layer of the framework is intentionality. In this context, we mean that a company’s business model should be purposefully designed around AI capability, rather just applying AI to existing processes. Spencer Fung, president and CEO of Li & Fung, a global supply chain and logistics company, gave us an analogy: “Companies acquiring AI without a new business model is like a company digitizing a horse and carriage—while the competition has created a digital automobile.” He offered the clothing company Shein as an exemplar: a company that became a fast-fashion industry leader by using a business model centered around a seamless e-commerce platform leveraging a data-driven approach.

Next comes integration across all functions of the enterprise, with horizontal communication and AI as the enabling layer—in other words, getting rid of silos. For example, ekaterra, the world’s largest tea company, invested in a unifying AI platform to create complete horizontal connectivity, integration, and real-time visibility.

The real challenge, however, is implementation. Garry Kasparov, former world chess champion, has written that winning performance does not come from combining the best technology with the best people—but from the best process of combining.⁴ To achieve this, talent must be familiar with AI capabilities and know how best to utilize them.

However, AI is an evolving technology and that necessitates a business add slack to the system to allow opportunity for learning. One of the authors learned this in an earlier project with Lockheed Martin on the F-22 Raptor, where downtime was needed for workers to become familiar with new tools and technologies.⁵ With an evolving technology new learning may even cause a temporary loss of productivity, but that loss will ultimately be more than made up—it's the proverbial, “putting down an axe to learn how to use a chain saw.”

Implementation also requires differentiation between “deep work” and “shallow work,” with the latter assigned to AI. Deep work, however, requires worker training for developing talent literacies. For example, one company builds in 90-minute sessions as part of the regular work week.

The last part of the framework is indication, or performance measurement. Forget traditional performance measures or the inane debate on working from office or home. Look to develop novel metrics tied directly to purposeful intentionality of your business model using generative AI. Organizations from Google to Schneider Electric are now using AI to redefine what to measure, how to measure, and how to improve performance.

Humans with Enhanced Skills “in the Loop”

Competitive advantage cannot be achieved without humans in the loop. Rushing to replace talent with AI is a huge mistake.

Why?

First, AI is copyable. What is not copyable is a unique business model, processes, and thoughtful integration of humans.

Second, AI is based on historical data that may not hold true in a volatile global business environment. Kinaxis's Sicard underscored that “every math-based model collapsed when the pandemic hit. None of the assumptive parameters could be trusted. It is not an indictment on the science ... but an indictment on the belief that those technologies eliminate the need to be agile.” Business decisions are not made in a vacuum separate from issues of labor, inflation, geopolitics—and this is where domain expertise of people is essential. In our executive interviews we repeatedly heard that the new competitive advantage is “the human mind,” “human critical thinking,” and “human decision intelligence.”

Third, AI is subject to hallucination and “drift,” where output is either fabricated by the AI or simply inaccurate. Handing the reins over to AI poses great risks in terms of inaccurate decision-making and even legal liability.

AI is still a tool. The centerpiece is people, but with enhanced human literacies, a well-thought-out business model, and superb processes that integrate humans with their AI copilots.

TAKEAWAYS

Leaders feel immense pressure to “do something” to implement AI solutions. But how do they actually integrate AI with their workforce to achieve good business outcomes? Companies should invest more in two important domains of human skills.

- # First are effective interpersonal skills, such as basic conflict resolution, communication, and mindfulness. These help foster team cooperation.

- # Second is domain expertise, with a focus on preserving knowledge among experienced talent and developing it among inexperienced workers. These skills help workers get the most out of AI tools—and make the best decisions when working with them.
- # While technical skills get more press, it is these uniquely human skills that companies need—and find to be in short supply in today's marketplace.

NOTES

1. Nada R. Sanders and John D. Wood, “The Secret to AI Is People,” hbr.org, August 24, 2020, <https://hbr.org/2020/08/the-secret-to-ai-is-people>.
2. Emily Field, Bryan Hancock, and Bill Schaninger, “Don’t Eliminate Your Middle Managers,” *Harvard Business Review*, July–August 2023, 48–52.
3. Sanders and Wood, “The Secret to AI Is People.”
4. Gary Kasparov, “Weak Human, Strong Force: Applying Advanced Chess to Military AI: War on the Rocks,” July 7, 2022, kasparov.com, www.kasparov.com/weak-human-strong-force-applying-advanced-chess-to-military-ai-war-on-the-rocks-july-7-2022/.
5. T. R. Browning and N. R. Sanders, “Can Innovation Be Lean?” *California Management Review* 54, no. 4 (2012): 5–19, <https://doi.org/10.1525/cmr.2012.54.4.5>.

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